



**Hybrid Human-AI Decision Support for Enhanced Human
Empowerment in Dynamic Situations**

D5.4 Training Centre Development for Human-AI Collaboration v2

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List of Abbreviations and Acronyms

Abbreviation	Meaning
EC	European Commission
EU	European Union
GA	Grant Agreement
HumAIne	Hybrid Human-AI Decision Support for Enhanced Human Empowerment in Dynamic Situations
WP	Work Package
LC	Learning Centre
LMS	Learning Management System

HumAlne consortium

The HumAlne consortium consists of the following organizations:

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1	COO	GFT	GFT ITALIA SRL	IT
2	BEN	IBM	IBM ISRAEL - SCIENCE AND TECHNOLOGY LTD	IL
3	BEN	Ebit	EBIT S.R.L.	IT
4	BEN	UPRC	UNIVERSITY OF PIRAEUS RESEARCH CENTER	EL
5	BEN	JSI	INSTITUT JOZEF STEFAN	SI
5.1	AE	QLE	QLECTOR, RAZVOJ CELOVITIH RESITEV ZA PAMETNE TOVARNE DOO	SI
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Executive Summary

The HumAIne project aims to advance effective and trustworthy Human-AI collaboration by combining technological innovation with strong human-centred design principles. A core component of this ambition is the **HumAIne Learning Centre**, conceived as a central digital hub for training, upskilling, and knowledge sharing, supporting the adoption and effective use of HumAIne technologies across diverse domains and user groups.

This deliverable, **D5.4 Training Centre Development for Human-AI Collaboration v2**, builds upon the foundations established in D5.3, which introduced the initial concept, structure, and requirements of the Learning Centre. While D5.3 focused on design, early implementation, and an initial training catalogue, the present deliverable reflects the **evolution** of the training strategy following the first phases of implementation, content development, and internal feedback within the project.

Since the submission of D5.3, the HumAIne consortium has reassessed its training approach based on early insights gained from content production, partner contributions, and the need to better address the heterogeneous profiles of future users. This process led to a structured redesign of the training offer, resulting in the definition of two complementary and differentiated training programmes: one targeting **technical users**, such as developers, integrators, and AI practitioners, and another addressing **non-technical users**, including workers, domain experts, and public-sector professionals involved in the project's pilots. These programmes are modular, role-oriented, and closely aligned with real-life use cases and deployment contexts.

In parallel, the Learning Centre has evolved into an operational digital environment that consolidates training materials, supports structured learning paths, and serves as the main access point for educational resources. Training materials, the Learning Centre, and live or recorded webinars are integrated as part of a coherent capacity-building ecosystem, where static content is complemented by interactive sessions and pilot-driven demonstrations (delivered via webinars). Content integration and expansion are carried out progressively, reflecting the Learning Centre's design as a living and reusable asset rather than a static repository.

Task 5.4 formally concludes at Month 27 of the project. However, training content development, platform population, and training activities using these assets, continue in close coordination with dissemination and exploitation actions in Work Package 7, ensuring sustained impact beyond this milestone. A dedicated white paper to be delivered later in the project will synthesise the complete training portfolio, lessons learned, and transferability potential of the HumAIne Learning Centre.

Overall, D5.4 documents the consolidation of a mature, feedback-informed training framework that strengthens HumAIne's capacity to empower users, foster trust in Human-AI collaboration, and support long-term adoption across sectors.

1 Introduction

1.1 Context and Scope

The HumAlne project addresses the challenge of enabling effective, trustworthy, and sustainable collaboration between humans and Artificial Intelligence (AI) systems in real-life, dynamic environments. While the project advances a range of technical innovations covering hybrid Human-AI decision-support architectures, collaborative AI paradigms, and orchestration mechanisms, it recognises that the successful adoption of such technologies depends equally on human understanding, skills, and trust.

Within this context, training and capacity-building activities play a central role in ensuring that HumAlne's technical results can be meaningfully applied across sectors and user profiles. Human-AI collaboration requires users not only to interact with AI-driven systems, but also to **understand their behaviour, limitations, and the role of human input** in shaping system outcomes. This is particularly relevant in operational contexts where decisions have real-world consequences, such as manufacturing, healthcare, energy, smart cities, and public services.

To address these needs, the HumAlne project has embedded training and learning activities as a core component of its impact pathway from the early stages. These activities aim to support a wide range of stakeholders, including **technical users** such as AI researchers, developers, and system integrators, as well as **non-technical users** such as domain professionals, workers involved in pilot deployments, policymakers, and the general workforce. The diversity of these audiences requires differentiated, accessible, and context-aware training approaches that go beyond generic AI literacy.

The **HumAlne Learning Centre** acts as the primary digital instrument through which this training strategy is implemented. It provides a structured environment for organising, accessing, and reusing training materials derived from the project's technical results, supporting learning paths that are aligned with user roles, knowledge levels, and real deployment scenarios.

1.2 From D5.3 to D5.4

Deliverable D5.3 presented the first version of the *Training Centre Development for Human-AI Collaboration*. Its primary focus was on defining the conceptual foundations of the HumAlne Learning Centre, including its objectives, target audiences, platform requirements, and an initial catalogue of training resources. At that stage, the emphasis was placed on design choices, early implementation steps, and the identification of learning needs based on the project's envisioned use cases.

The **present deliverable, D5.4**, builds directly upon this foundation. It does not represent a change in direction, but rather a natural evolution of the Learning Centre and training strategy as the project progressed from conceptualisation to implementation and consolidation. As training materials began to be produced and integrated, and as partners contributed content linked to their respective technical developments and pilots, valuable insights emerged regarding audience needs, content structure, and effective learning formats.

These insights highlighted the need for a more explicit differentiation between technical and non-technical users, clearer learning pathways, and stronger alignment between training content and the operational realities of the HumAlne pilots. D5.4 captures this evolution by documenting how the

initial training approach was refined, how training programmes were restructured, and how the Learning Centre has been consolidated as an enabling infrastructure for capacity building.

1.3 Deliverable Objectives

The objectives of D5.4 are to:

- Document the development and consolidation of the HumAlne Learning Centre under Task 5.4.
- Describe the evolution of the training methodology following early implementation experience and internal feedback.
- Present the structure and rationale of the training programmes developed for technical and non-technical users.
- Explain how training materials derived from HumAlne's technical results are integrated into a coherent learning framework.
- Clarify the scope of Task 5.4 in relation to other project activities, particularly the distinction between training asset development and training activity execution.

By addressing these objectives, the deliverable provides a transparent and up-to-date account of how the HumAlne project supports learning, upskilling, and user empowerment in the context of Human-AI collaboration.

1.4 Scope, Timing, and Task Framing

Task 5.4 Training Programs and Learning Center Development spans from M4-M27 of the HumAlne project. Its scope covers the development of the Learning Centre and the associated training programmes, including the integration of content and lessons derived from the project's technical tasks. Specifically, T5.4 is responsible for designing and consolidating training programmes for both technical and non-technical users, developing multimedia training materials, and integrating these assets into a structured digital Learning Centre.

The outputs of T5.4 include training materials in multiple formats (e.g., videos, presentations, visual resources, and technical documentation) that allow users to familiarise themselves with the HumAlne platform, its operating system, collaborative AI technologies, and other related materials. These assets are designed to enable learning, understanding and knowledge transfer, and to support the adoption of HumAlne results across different domains.

It is important to distinguish between the development of training assets and the implementation of training activities. While T5.4 delivers the training programmes, materials, and Learning Centre infrastructure, the execution of concrete training activities, such as live webinars, workshops, or other dissemination-oriented events making use of these assets, is addressed under separate tasks within the project. This separation ensures a clear allocation of responsibilities between the creation of reusable training resources and their deployment within broader engagement and dissemination actions.

T5.4 formally concludes at M27. At this milestone, the Learning Centre is operational and the training framework is consolidated. At the same time, the Learning Centre has been intentionally designed as a living and reusable asset. Additional training materials continue to be refined and integrated in coordination with other work packages, reflecting a feedback-driven and iterative approach rather than a static delivery model. A dedicated white paper to be produced later in the project will synthesise

the full training portfolio, lessons learned, and the broader transferability potential of the HumAlne Learning Centre beyond the project's lifetime.

1.5 Structure of the Deliverable

This deliverable is structured to reflect the evolution, consolidation, and current status of the HumAlne Learning Centre and training programmes developed under T5.4. Each section addresses a specific aspect of the training strategy, moving from conceptual evolution to concrete implementation and forward-looking perspectives.

- Section 1 provides an introduction to this deliverable and sets the scene for the rest of the document. It includes some comments about the previous version of the deliverable, states the objectives, and conceptualizes the task reported on.
- Section 2 describes the methodological evolution of the training approach and feedback-driven refinement of the materials.
- Section 3 shares an updated view of the Learning Centre's target audience based on the new training programmes.
- Section 4 introduces the overall structure of the HumAlne training offer, split into two main training programmes.
- Section 5 discusses the concrete training content available at M27, described in a progressive and modular structure.
- Section 6 describes the Learning Centre's format, structure, and consolidation based on received feedback.
- Section 7 outlines some forward-looking perspectives within the broader project timeline, including future additions and the contribution of a whitepaper, and, as the final section, summarizes the key achievements of T5.4 and reflects on the contributions of the Learning Centre to the project's overall objectives.

2 Methodological Evolution and Feedback Driven Refinement

The development of the HumAlne Learning Centre and associated training programmes followed an iterative and evidence-informed methodology. While D5.3 established the initial conceptual framework and training approach, subsequent implementation activities, content development efforts, and internal feedback highlighted the need for refinement. This section documents how the training methodology evolved between the first and second versions of the deliverable, leading to a more structured and content-aware training framework.

2.1 Initial Assumptions (as per D5.3)

In its initial formulation, as described in D5.3, the HumAlne training approach was based on a unified Learning Centre offering a broad catalogue of training resources addressing multiple aspects of Human-AI collaboration (swarm learning, EU AI Act, etc.). The underlying assumption was that a single, comprehensive catalogue could serve the needs of a wide and heterogeneous audience. This initial methodology prioritised:

- The establishment of a common Learning Centre as a central access point for training materials.
- The integration of diverse content formats, including self-paced courses, technical documentation, and introductory learning resources.
- A relatively uniform treatment of target audiences, with limited differentiation in content depth or learning pathways between technical and non-technical users.

This approach was appropriate for the early stages of the project, when the primary objective was to define the Learning Centre's scope, validate its relevance, and initiate content production. However, as training materials began to be developed and integrated into the Learning Centre, limitations of this model became apparent.

2.2 Feedback Sources and Insights

Feedback informing the methodological refinement was gathered through multiple channels during the implementation phase of Task 5.4. These included internal reviews of draft training materials, exchanges with technical partners contributing content, and reflections emerging from the process of structuring and integrating materials into the Learning Centre. Several important insights emerged from this process.

- Firstly, **user needs are not homogeneous**. Technical users (e.g., AI developers, system integrators, data scientists) and non-technical users (e.g., front-line workers, municipal officers, domain-specific professionals) have fundamentally different expectations regarding the content depth, terminology used, and learning objectives of training material.
- Next, it is necessary to consider the **cognitive load and accessibility** of the materials. The content meant to capture the full technical complexity of HumAlne systems risked overwhelming non-technical audiences, while more introductory or high-level resources did not sufficiently support advanced technical users.
- There is also a **need for clearer learning pathways** in the HumAlne training programme. The initial catalogue made it difficult to guide users through a coherent learning progression aligned with their role, prior knowledge, and intended use of the platform.

- Feedback collected pointed to a further **importance of pilot alignment** in order to highlight the value of contextualised, use-case driven content that connects abstract Human-AI concepts to concrete operational scenarios.

Overall, these insights did not challenge the relevance of the Learning Centre itself, but rather pointed to the need for a more explicit methodological structure underpinning the training offer.

2.3 Resulting Strategic Adjustments

In response to the feedback collected, T5.4 implemented a set of strategic methodological adjustments to the training approach.

The most significant adjustment was the transition from a single, undifferentiated training catalogue to **two distinct and complementary training programmes**, targeting technical and non-technical users respectively. This programme-based structure enables clearer learning objectives, more appropriate content depth, and tailored learning pathways for different user profiles, while maintaining a shared conceptual foundation around Human-AI collaboration.

Additional refinements included:

- **Modularisation** of training content, with clearly defined scopes and learning outcomes for each module, facilitating progressive learning and reuse across domains.
- Stronger **alignment with technical results and pilots**, ensuring that training materials are grounded in concrete HumAlne components, architectures, and deployment experiences.
- Clear **separation between asset development and activity execution**, whereby T5.4 focuses on designing and integrating training materials and Learning Centre infrastructure, while training activities making use of these assets are implemented under other tasks.
- Inclusion of a more **systematic approach to monitoring and evaluation** in order to assess the engagement, effectiveness, and reach of the Learning Centre moving forward.

These methodological adjustments resulted in a more mature and robust training framework that remains faithful to the original objectives of D5.3 while better addressing the practical realities of Human-AI collaboration, user diversity, and long-term sustainability. The refined approach positions the HumAlne Learning Centre as a flexible yet structured capacity-building asset capable of evolving alongside the project and beyond its duration.

3 Updated Target Audience and Learner Personas

The redesign of the training strategy under T5.4 required a more precise and structured understanding of the audiences who will interact with the HumAlne Learning Centre and the associated training programmes. While D5.3 outlined a broad audience landscape, ranging from researchers to policymakers, implementation experience and feedback demonstrated that a refined categorisation was necessary in order to support clearer learning pathways and more appropriate depth of content.

This section, therefore, discusses the updated audience model adopted for the Learning Centre and introduces a set of representative “learner personas.” The personas support the user-centred design approach behind the training programmes, and ensure that content development remained aligned with the realities and needs of end users across sectors and roles.

3.1 Refined Audience Model

The audience model for the Learning Centre is now structured around **two principal user groups**: technical and non-technical users. This division reflects the substantial differences in learning needs, expected interaction with the HumAlne platform, and familiarity with AI concepts and methods.

Technical Users are individuals who engage directly with the development, configuration, analysis, and integration of AI-driven systems. Their work relies on an understanding of complex technical components within the HumAlne platform, including deployment infrastructures, orchestration mechanisms, Active Learning modules, benchmarking environments, and explainability frameworks. Examples include AI developers, data scientists, system integrators, and research scientists. For these users, the Learning Centre must provide access to detailed documentation, platform architecture explanations, hands-on demonstrations, and advanced technical modules.

Non-technical users, on the other hand, are individuals whose role involves applying the platform’s functionalities rather than developing or configuring them. This category includes operators and workers in the project pilots, decision-makers and managers, public-sector professionals, and potential future adopters in operational settings. Their priorities centre around understanding Human-AI collaboration concepts in plain language, engaging with interface elements, interpreting outputs safely, navigating the platform, and understanding regulatory and ethical considerations as they relate to responsible use of AI systems.

This refined audience structure addresses a key challenge identified during the early content development phase: attempting to serve both groups with identical material risked compromising usability for all users. On one hand, the technical users required depth and precise detail, while on the other hand, non-technical users needed accessible, curated learning resources to familiarize themselves with key concepts, terminology, and the platform. The two-programme model enables the Learning Centre to clearly align its content with user needs, ensuring a positive training offer and experience for all stakeholders.

Moreover, this audience division ensures that the broader training objectives of the project are supported. Technical users will be able to meaningfully engage with the platform and integrate its components into new or existing environments, and non-technical users will be empowered to interact with AI-assisted systems in real-world use case scenarios. Both user groups benefit from shared conceptual foundations behind Human-AI collaboration (trust, ethics, EU AI Act), and now through the dual training programmes, they also have tailored content for their respective roles and contexts.

3.2 Learner Personas

To translate the audience model into a more tangible guide for training content, a set of representative learner personas has been developed. These personas do not describe individual users but rather illustrate common profiles of potential Learning Centre users, including motivations, learning requirements across the project’s target groups. They serve as orientation points for training programme construction and to help ensure materials respond effectively to user needs. First the technical user learner personas are presented (T1, T2 and T3), followed by the non-technical user learner personas (N1, N2 and N3).

3.2.1 Persona T1 – “The AI Developer”

The AI Developer persona represents highly technical users deeply engaged with the computational and algorithmic aspects of the HumAIne platform. These learners seek content that moves beyond conceptual explanation and provides direct access to tools, code demonstrations, APIs, deployment guides, and benchmarking frameworks. They are motivated by opportunities to use HumAIne modules to develop new research outputs or product-level features. For this audience, they can benefit from structured walkthroughs that enable them to test and evaluate workflows. Their challenges revolve around locating sufficiently detailed documentation and understanding the interactions between platform components. The learning programme therefore aims to guide AI Developers through deeper modules.

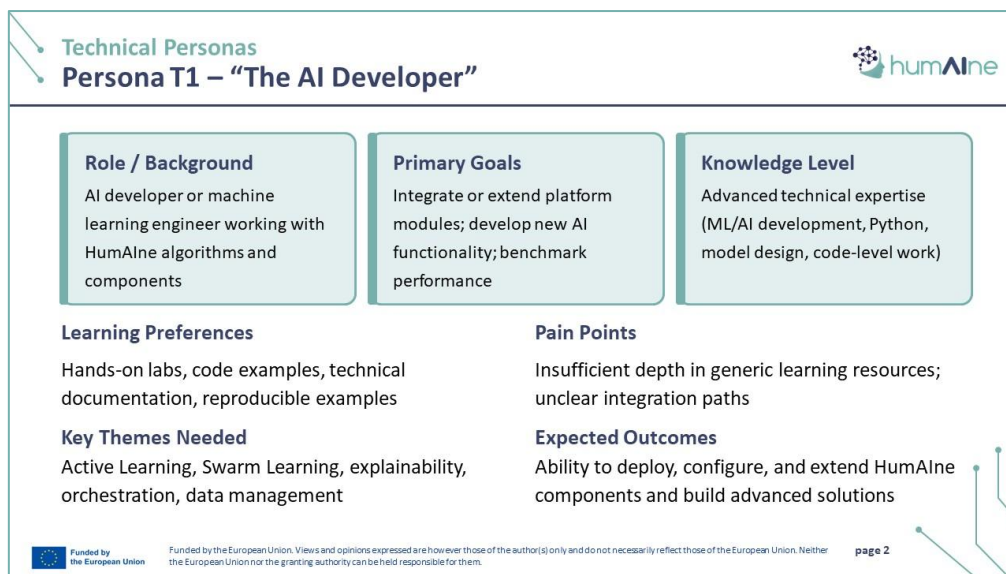


Figure 1: Learner Persona T1 - "The AI Developer"

3.2.2 Persona T2 – “The Systems Integrator”

This persona is responsible for embedding HumAIne technologies within real operational environments, such as industrial plants, hospitals, or municipal systems. Unlike AI Developers, these individuals require a deep understanding of system-level behavior, data flows, infrastructure compatibility, and orchestration pipelines. Their learning journey prioritises practical deployment guidance, environment configuration steps, interface navigation, and architecture-level understanding. They value learning resources that clarify technical dependencies, workflows, and integration troubleshooting. For these users, the training programme is designed around platform

architecture and orchestration layers to ensure that deployments are secure, scalable, and interoperable.

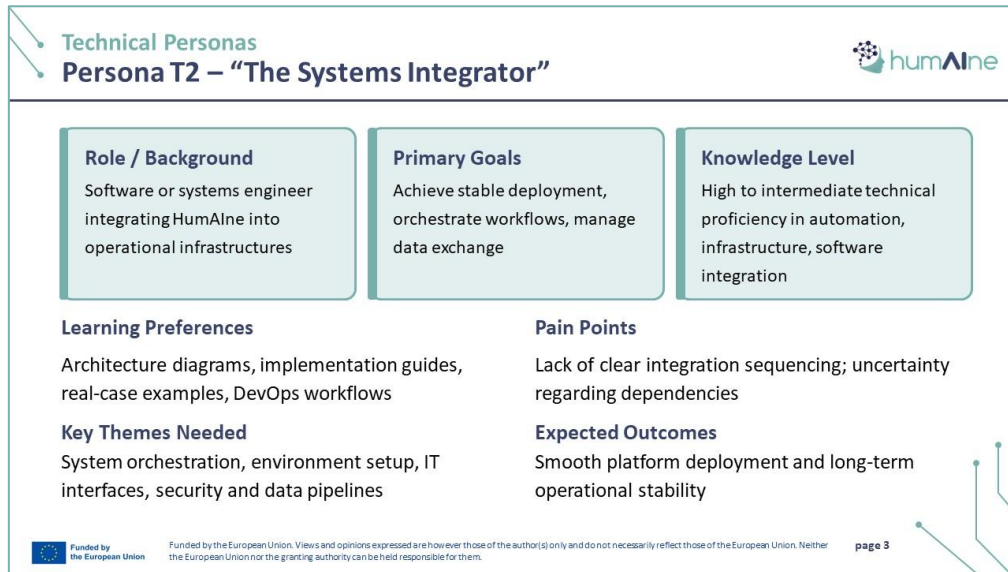


Figure 2: Learner Persona T2 - "The Systems Integrator"

3.2.3 Persona T3 – “The Research Scientist”

The Research Scientist persona is motivated by understanding how HumAIne supports experimentation and advances the science behind Human-AI Collaboration. Their training materials must include conceptual models and frameworks, sociotechnical integration points, and comparative evaluation strategies. These learners value modelling explanations, pilot analysis, and documentation that supports research reproducibility. The programme for this persona prioritises conceptual depth and methodological clarity rather than hands-on deployment, enabling scientists to better understand and use HumAIne’s academic results.

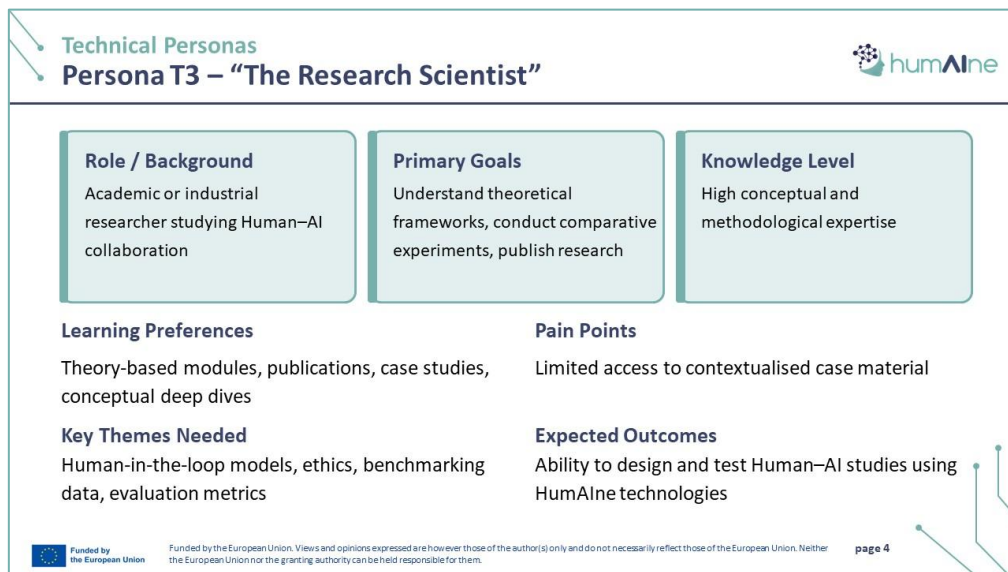


Figure 3: Learner Persona T3 - "The Research Scientist"

3.2.4 Persona N1 – “The Pilot Worker / Operator”

The Pilot Worker persona represents all the frontline operators who directly interact with the AI system through visual interfaces or decision-support tools. They may operate in the same domains as HumAIne’s pilot use cases, or in others beyond the pilots’ scope. They do not require algorithmic understanding; instead, they need simple, direct guidance explaining what the platform does, why it matters, and how to interpret outputs. Training therefore prioritises accessibility, simplicity, and reassurance. The goal is to reduce cognitive load, build confidence, and ensure safe interaction. Their content must avoid technical jargon and prioritise high-impact demonstrations and scenario examples connected to the various use cases of the pilot.

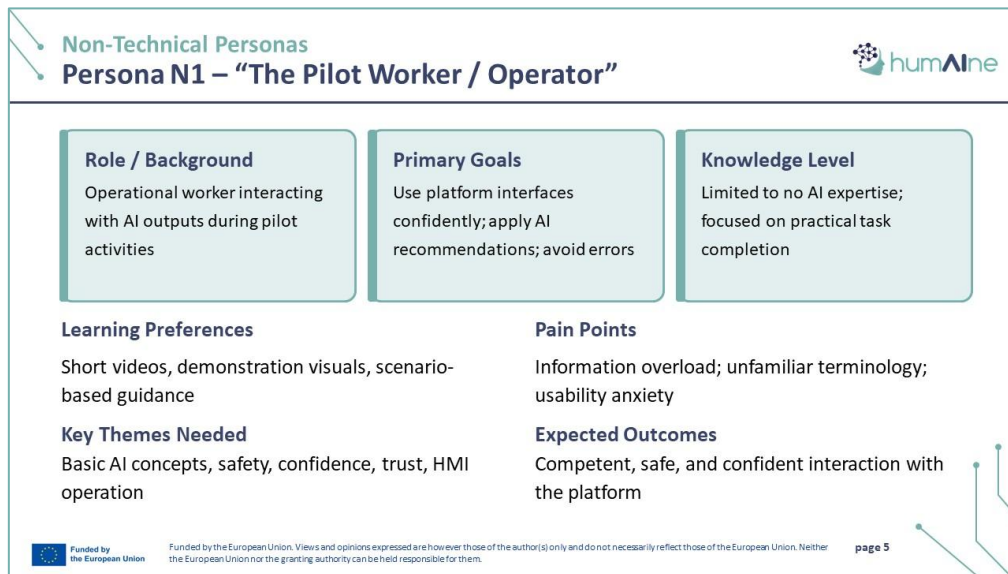


Figure 4: Learner Persona N1 - "The Pilot Worker / Operator"

3.2.5 Persona N2 – “The Decision-Maker / Manager”

The persona of the Decision-Maker or Manager is responsible for interpreting AI-supported information and shaping operational decisions. Their learning materials highlight business value, transparency, and organisational benefits rather than technical training. They need clarity on how Human-AI collaboration improves efficiency and quality, how outputs should be interpreted, and how accountability and oversight function. Their training pathway focuses on strategic understanding and platform literacy rather than deep immersion.

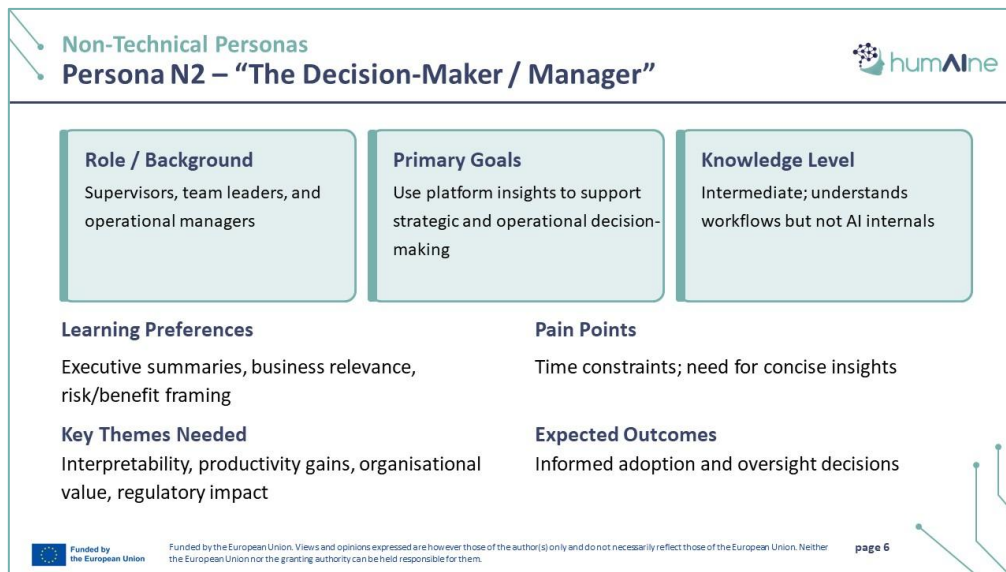


Figure 5: Learner Persona N2 - "The Decision-Maker / Manager"

3.2.6 Persona N3 – “The Policymaker / Public-Sector Professional”

The Policymaker persona focuses on legal, ethical, and societal implications rather than operational mechanics. Their training framework highlights safeguards, transparency, human oversight expectations, and alignment with European AI governance structures. Accessible communication and conceptual clarity are essential, and content is framed to facilitate better understanding for policymaking and responsible adoption.

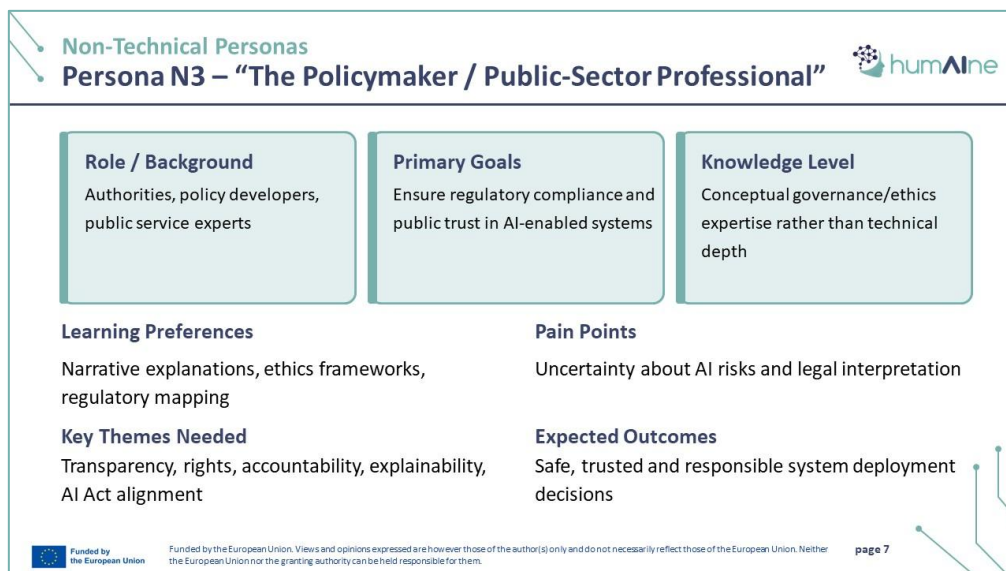


Figure 6: Learner Persona N3 - "The Policymaker / Public-Sector Professional"

As a whole, these personas demonstrate the heterogeneous needs of users within the HumAIne ecosystem, highlighting why differentiated training programmes and materials were necessary, why modularisation and content depth vary across courses, and why the Learning Centre incorporates a wide range of formats. These personas also provide a foundation for ongoing refinement and future training activities that will take place in WP7.

4 Training Programme Structure

The training programmes developed under T5.4 are designed to support learning needs that are diverse in depth and content expectations. Building on the methodological refinement described in Section 2 and the persona framework described in Section 3, the HumAlne training offer is now structured into two distinct but complementary training programmes: one targeting **technical users** and one targeting **non-technical users**. These programmes are organised into modular pathways to support gradual competence development and usability across different types of stakeholders.

This structure enables the LC to offer a coherent and navigable training format, positioning its training resources in a way that reflects learner goals, professional roles, and interaction with the HumAlne platform. Both programmes utilise multimedia content, documentation, visual learning material and other guides while remaining differentiated in depth, technical detail, and practical uses.

4.1 Overview of the Training Programmes

The HumAlne training offer consists of two curated and role-oriented training programmes:

- **Technical Users Training Programme**, aimed at profiles involved in AI development, integration, research, and system orchestration.
- **Non-Technical Users Training Programme**, designed for workers, domain professionals, managers, policymakers, and citizens who interact with the platform outputs or participate in decision-making processes involving Human-AI collaboration systems.

Each programme is organised into clearly defined learning modules, supported by multimedia assets including recorded lectures, presentations, explanatory documents and written guides, and demonstrations (either pre-recorded or in the form of upcoming webinars). The modular structure allows users to select content at appropriate depth levels for their profiles, progress through recommended pathways, and revisit resources in alignment with their professional needs.

The programmes connect conceptual orientation with practical insights. For technical users, the training provides detailed insight into the platform's architecture and operational components, including algorithmic design and orchestration workflows. For non-technical users, the training emphasizes familiarity, confidence, interpretation of outputs, and an understanding of how Human-AI collaboration applies in real operational settings.

4.2 Technical Users Training Programme

The Technical Users Training Programme, entitled, "Engineering Human-AI Systems with the HumAlne Platform," is designed to provide in-depth technological and architectural knowledge for users who directly engage with the development, configuration, and integration of HumAlne platform components. This programme targets advanced professional profiles (e.g., AI developers, data scientists, research engineers, system integrators, technical pilot contributors), responding to the specialised needs identified through feedback evaluation and persona analysis.

The programme enables technical users to understand and interact with the experimental, computational, and infrastructural aspects of the HumAlne ecosystem, and offers a structured learning pathway that combines conceptual knowledge, system deployment guidance, hands-on methods,

advanced AI techniques, human-machine interface considerations, benchmarking strategies, and regulatory frameworks.

The programme is delivered online through a combination of learning assets, including presentations and recorded lectures, downloadable guides, and technical documentation, and it will be updated periodically to include additional whitepapers, deliverables, or webinar recordings that can further strengthen the programme. These diverse formats accommodate varied learning preferences and allow participants to revisit the content according to their needs.

“Engineering Human-AI Systems with the HumAlne Platform” reflects the technological innovations of the HumAlne project and prepares participants to work with the platform components, including the experimental environments, orchestration layers, and human-centred AI methodologies. It supports the long-term ambition of building capacity among research and industrial communities via the skills required to extend, adapt, and deploy Human-AI collaborative systems.

4.2.1 Programme Architecture

The programme architecture for technical users is structured into **seven dedicated modules**, progressing from conceptual foundations to advanced technological implementation and sector-based application. Modules are designed to operate independently while remaining interconnected within a coherent progression structure. Each module includes learning objectives, specialised topics, associated digital assets, and, where relevant, webinar delivery formats.

Below is an outline of the seven modules:

4.2.1.1 Module 1: Foundations of Human-AI Collaboration

Human-AI collaboration represents the conceptual foundation of the entire HumAlne training offer. Module 1 introduces the core principles and paradigms that define how human decision-making integrates with intelligent systems. It provides participants with a clear understanding of the Human-in-the-Loop (HITL) and Human-in-Command (HIC) approaches, emphasising why real-time human judgement remains essential for safe and accountable AI behaviour.

The module situates collaboration within a broader sociotechnical frame, showing how modern AI workflows require humans to supervise, interpret, validate, and guide automated recommendations. This includes the role of co-creation and participatory design in shaping system interfaces, ensuring usability and trust, and preventing unwanted automation bias. Through conceptual examples drawn from HumAlne pilot domains (smart cities, medical applications, industrial operations), learners gain an appreciation for how collaboration changes daily workflows and why it enables higher quality decision-making than purely autonomous approaches.

The module also provides the theoretical grounding that supports later modules, particularly those on deployment, evaluation, and interaction. By completing Module 1, participants will be ready to understand why HumAlne tools and methods are structured around continuous feedback cycles, explainability requirements, and shared responsibility frameworks.

Table 1: Overview of Technical Users' Programme Module 1.

Component	Details
Objective	To provide a robust conceptual framework around Human-in-the-Loop (HITL) and Human-in-Command (HIC) paradigms, empowering developers and

	system engineers to implement AI systems that prioritise safety, usability, and ethical co-decision-making.
Topics/Lessons	• Key principles of Human-AI collaboration • HITL and HIC definitions, success factors, and real-world examples • Co-creative and participatory design methodologies • The HumAlne vision: positioning the human at the centre
Delivery Formats	Online recorded presentations and lecture videos; downloadable slide decks; written supporting material (project deliverables) and conceptual reading resources.

4.2.1.2 Module 2: HumAlne Platform: Deployment and Customisation

Module 2 provides the first practical and technical entry point into the HumAlne platform and its operational environment. It introduces the platform architecture, deployment workflow, orchestration stack, security foundations, and DevOps processes required to run HumAlne in real-world infrastructures. Through detailed architectural guidance, participants learn how the platform is installed, configured, integrated, and maintained across distributed environments.

This module draws directly on technical material from the HumAlne Architecture lecture and the GFT Factory deployment resources, offering a concrete view of container orchestration using Kubernetes, automated CI/CD pipelines, code delivery management, and full-stack configuration principles. The module enables participants to understand how components interconnect, how APIs expose platform logic to external systems, and how orchestration pipelines ensure repeatable deployment and scaling.

The training highlights good practices for secure and compliant operation, including data handling awareness and infrastructure trust requirements. It supports developers, integrators, and engineers in progressing from theoretical platform understanding to hands-on implementation knowledge. Module 2 therefore establishes a bridge between conceptual foundations and the more advanced AI technique modules that follow, ensuring that learners can deploy the platform components and environments required to test, refine, and operate HumAlne technologies.

Table 2: Overview of Technical Users' Programme Module 2.

Component	Details
Objective	To train users in the setup, deployment, and full-stack configuration of the HumAlne platform, including orchestration workflows, API integration, and DevOps practices in secure environments.
Topics/Lessons	• HumAlne architecture overview • Platform bootstrapping and deployment scenarios • Orchestration techniques and DevOps practices in GFT infrastructure • API layers and external integration points • Security, trust, and compliance in orchestration pipelines
Delivery Formats	Online recorded presentations and lecture videos; downloadable slide decks; platform documentation; written supporting material (project deliverables) and future webinar recordings.

4.2.1.3 Module 3: Advanced AI Techniques for Collaborative Systems

Module 3 extends the programme into advanced algorithmic and conceptual methods underpinning HumAlne’s human-centred AI architecture. It provides participants with a structured overview of four major collaborative AI technologies (“components”) integrated into the project: Active Learning, Swarm Learning, Neuro-Symbolic AI, and Explainable AI (XAI) methodologies. The module connects theory to implementation by demonstrating how these approaches enhance learning efficiency, adaptability, federated intelligence, and reasoning transparency in Human-AI workflows.

Participants explore how decentralised and privacy-preserving collaboration is achieved through Swarm Learning and how human feedback supports Active Learning and iterative improvement. Neuro-Symbolic AI material introduces the integration of machine learning with symbolic reasoning to enhance interpretability and decision generalisation. The module also introduces widely used explainability techniques, establishing a core framework for transparency and debugging in later deployment tasks.

The learning material is supported by dedicated study guides and recorded webinars, with additional webinars scheduled across 2026 for more detailed demonstrations of these components.

Table 3: Overview of Technical Users' Programme Module 3.

Component	Details
Objective	To explore and apply advanced AI methods integrated in the HumAlne platform, Active Learning, Swarm Learning, Neuro-Symbolic AI, and XAI, with emphasis on iterative human feedback and system improvement.
Topics/Lessons	- Fundamentals of Active Learning (human supervision) - Swarm Learning for decentralised collective intelligence - Neuro-symbolic architectures for reasoning and adaptability - Explainability techniques (LIME, SHAP, PDP) - Application cases in various sectors (smart cities, manufacturing, healthcare)
Delivery Formats	Online recorded presentations and lecture videos; downloadable slide decks; written supporting material (project deliverables) and future webinar recordings. Scientific publications will be linked, where available.

4.2.1.4 Module 4: Interfaces for Human-AI Interaction

Module 4 introduces participants to the visual, multimodal, and immersive interface technologies developed within HumAlne to support Human-AI co-decision making. It presents the architectural and interaction design principles behind the Human-Machine Interface (HMI) systems deployed in project pilots, focusing on clarity, ergonomics, responsiveness, and situational awareness.

Participants are introduced to multiple interface categories developed across the platform, including conversational interfaces, dashboard-based control layers, and Mixed Reality (MR) or Augmented Reality (AR) decision-support environments. The training explains how these interfaces translate complex AI outputs into human-accessible visual representations, how visualisation logic supports trust and safe decision adoption, and how interface prototyping can be used to evaluate usability and workflow fit.

Through this module, learners gain insight into how interface architectures vary across pilot scenarios. The module introduces best-practice design thinking, demonstrating how user feedback and ergonomics shape the evolution of interface implementation within HumAIne.

Table 4: Overview of Technical Users' Programme Module 4.

Component	Details
Objective	To enable developers and integrators to design ergonomic, multimodal interfaces that support human-centred collaboration with AI in complex operational environments.
Topics/Lessons	• HMI design principles: clarity, ergonomics, responsiveness • AR and MR use cases • Visualisation frameworks for AI decision feedback • Interface prototyping and usability evaluation
Delivery Formats	Downloadable slide decks; technical descriptions; written supporting material (project deliverables) and future webinar recordings.

4.2.1.5 Module 5: Benchmarking and Performance Evaluation

Module 5 focuses on the measurement and evaluation of Human-AI collaboration and system-level performance. It introduces participants to the HumAIne benchmarking suite and its associated metrics, which go beyond classical model accuracy to include trust, usability, fairness, human effort, efficiency, and interaction-based indicators.

Participants learn how to interpret system performance using structured industrial and human-centric KPIs, understand how dashboards visualise results, and apply benchmarking methods to assess iterative improvements. The module explains how evaluation logic is embedded into the platform, including logging workflows, performance data extraction, and comparative system analysis. It also provides insight into how performance indicators are applied to pilot settings and simulation environments to validate human-machine interaction outcomes.

The training prepares learners to integrate benchmarking into development and deployment workflows, supporting continuous improvement and improving system accountability. This competency enables technical users to align development practices with operational performance expectations and human collaboration requirements.

Table 5: Overview of Technical Users' Programme Module 5.

Component	Details
Objective	To introduce the HumAIne benchmarking suite and guide participants in evaluating technical and industrial KPIs in the context of human-AI collaboration.
Topics/Lessons	• KPIs for trust, accuracy, fairness, and usability • Technical validation of AI models • Human-centred performance metrics • Using benchmarking dashboards for continuous improvement
Delivery Formats	Benchmarking dashboards; technical manuals; downloadable learning materials; webinar recordings.

4.2.1.6 Module 6: Regulation and Ethical Compliance

Module 6 delivers essential training on ethical and regulatory frameworks governing trustworthy AI. It explains the principles, legal obligations, and compliance processes defined by the EU AI Act and connects them to HumAlne platform development workflows. The module guides participants in understanding system risk categories, transparency requirements, governance procedures, and traceability processes shaping responsible AI development and deployment.

Learners are introduced to fairness, accountability, and transparency concepts and gain insight into bias management and responsible data handling. Through this material, technical users gain the ability to align platform development with policy expectations and ensure that system design remains safe, lawful, and ethically grounded.

Module 6 also contextualises how compliance emerges across different pilot environments, demonstrating how ethical considerations support operational responsibility, human dignity, and the long-term social acceptance of AI technologies.

Table 6: Overview of Technical Users' Programme Module 6.

Component	Details
Objective	To ensure participants can align development practices with EU regulatory frameworks, especially the AI Act, and embed ethics-by-design in hybrid systems.
Topics/Lessons	• Overview of EU AI Act obligations for developers • Trustworthy AI principles: transparency, accountability, fairness • Ethical risks in hybrid AI systems • Audit trails, bias monitoring, and governance mechanisms
Delivery Formats	Study guides; online recorded presentations and lecture videos; downloadable slide decks; written supporting material and conceptual reading resources.

4.2.1.7 Module 7: Pilot Applications and Sectoral Implementations

Module 7 brings together all preceding technological and conceptual content by demonstrating how the HumAlne platform operates within real deployment settings. The module introduces participants to project pilot environments and use cases, highlighting technical goals, integration strategies, and domain-specific challenges.

It showcases how HumAlne technologies adapt across manufacturing optimisation tasks, smart city workflow automation, healthcare support systems, ticketing services, and energy network management. Participants examine the practical conditions under which platform components are deployed, gaining insight into scalability needs, data integration requirements, workflow adaptation, and human collaboration opportunities.

This module will be delivered primarily through interactive pilot-focused webinars scheduled across 2026, providing extensive opportunities for knowledge transfer, discussion, and case reflection. By the

end of this module, learners will understand how the HumAlne platform translates into operational value and will be able to draw lessons for future deployments.

Table 7: Overview of Technical Users' Programme Module 7.

Component	Details
Objective	To explore the real-world deployment of the HumAlne platform across six pilot cases, identifying technical lessons and adaptation strategies.
Topics/Lessons	According to HumAlne pilot use cases.
Delivery Formats	Webinars (2026); technical documentation; project deliverables.

4.3 Non-Technical Users Training Programme

The Non-Technical Users Training Programme, entitled, “**Empowering Workers in Human-AI Collaboration**” is designed to provide accessible, workplace-oriented learning for individuals who interact with the outcomes and interfaces of the HumAlne platform rather than its internal technical components. The programme targets a broad range of end users (e.g., frontline operators, domain professionals, managers, supervisors, municipality workers, and other non-technical individuals) involved in Human-AI collaboration in diverse environments.

The programme focuses on building conceptual understanding, trust formation, confidence in system interaction, and an ability to interpret AI outputs responsibly in workplace settings. It prioritises plain language explanations and avoids complexity while introducing essential principles of Human-AI collaboration, system interpretation, and responsible use.

Like the previous programme, training is delivered online through accessible multimedia formats, including short learning videos, recorded lectures and presentations, downloadable guides and documentation, and webinars.

This programme plays a critical role in enabling the wider adoption of the HumAlne platform. By fostering comprehension, user trust, and safe interaction behaviours, “**Empowering Workers in Human-AI Collaboration**” supports technical acceptance and long-term sustainability. The training modules provide workers with concrete examples of how Human-AI collaboration improves work quality, strengthens productivity, reduces workload, and enhances decision-making, all while ensuring an ethical and responsible use of AI-based systems.

4.3.1 Programme Architecture

The programme architecture for non-technical users consists of **six modules**, progressing from foundational literacy to workplace interaction, sector demonstration, and regulatory awareness. The modules can be followed sequentially or independently according to each user’s needs and workplace requirements. Each module contains a clear objective and lesson structure.

Below is an outline of the six modules:

4.3.1.1 Module 1: Human-AI Collaboration Essentials

Module 1 introduces non-technical participants to the core purpose, meaning, and function of Human-AI collaboration in contemporary workplace environments. It explains how AI systems support, rather

than replace, human judgement and highlights why collaboration between humans and intelligent systems can lead to better outcomes in productivity, accuracy, and decision-making.

The module also builds confidence and trust among learners by addressing basic definitions and demystifying terminology commonly associated with AI. It clarifies what AI is, how it operates at a conceptual level, and what distinguishes modern AI technologies from traditional software tools. Through relatable narrative examples (e.g., manufacturing workflows, healthcare assistance scenarios, smart infrastructure management), participants learn how Human-AI collaboration appears in real practice and what benefits it can bring to their daily activities.

By completing Module 1, participants will establish a foundational understanding needed to progress throughout the programme, gaining clarity on core vocabulary, expectations, opportunities, and potential risks.

Table 8: Overview of Non-Technical Users' Programme Module 1.

Component	Details
Objective	To build foundational understanding of how humans and AI systems collaborate, and why this approach enhances work quality and decision-making.
Topics/Lessons	• What Human-AI collaboration is and why it matters • Key terms and misconceptions • Examples from real work environments (e.g., factories, hospitals, energy control rooms) • Benefits and risks of AI-assisted work
Delivery Formats	Short presentations; recordings.

4.3.1.2 Module 2: AI Concepts for Daily Use

Module 2 expands participants' understanding by presenting core AI concepts in clear, accessible language. Rather than focusing on technical construction, learners explore how AI appears in everyday activities, why AI makes suggestions, and how intelligent systems adapt based on human input.

The module explains the difference between traditional software and AI-powered systems, emphasising that modern AI operates dynamically: learning from data, user interactions, and contextual feedback rather than executing fixed rules. Participants are also introduced to concepts such as machine recommendations, personalisation, and the role of user feedback in improving system performance.

Through this practical framing, participants develop a strong awareness of why their interactions matter and how their decisions, criticisms, and corrections influence the overall behaviour and accuracy of AI tools. This prepares them to engage confidently with AI-supported workflows and, later, with system interpretation and feedback reporting.

Table 9: Overview of Non-Technical Users' Programme Module 2.

Component	Details
Objective	To demystify the main AI mechanisms used in HumAIne, including Active Learning and Explainable AI, so users can interact confidently with the system.

Topics/Lessons	• AI explained in everyday language • Why user feedback improves AI systems • Understanding system suggestions • Differences between fixed software and intelligent platforms
Delivery Formats	Short videos; downloadable guides; introductory slide material; interactive short quizzes or user reflection activities.

4.3.1.3 Module 3: Trusting and Understanding AI Systems

Module 3 focuses on developing safe, informed, and confident interaction practices with AI systems. It addresses one of the central concerns in Human-AI collaboration: knowing when to rely on machine outputs and when to question them. Participants learn how to interpret system recommendations responsibly, understand limitations, and identify unusual or unexpected behaviours.

The module highlights that AI outputs are probabilistic, not absolute, and that human supervision remains essential. It also trains users on how to report errors, anomalies, or confusing results in a structured and constructive way once system reporting workflows are integrated within the HumAlne platform. Guidance is provided on recognising when something may be wrong and documenting findings clearly.

Module 3 therefore equips non-technical users with a practical troubleshooting mindset, enabling them to participate actively in improving Human–AI collaboration outcomes.

Table 10: Overview of Non-Technical Users' Programme Module 3.

Component	Details
Objective	To help workers interpret outputs and act safely when collaborating with AI, fostering trust through transparency and basic troubleshooting.
Topics/Lessons	• Interpreting model recommendations • Recognising system limitations • How to report problems or unexpected behaviours • Practical guidelines for safe interaction
Delivery Formats	Short videos; reporting and troubleshooting explanation slides; workplace demonstration content; scenario-based examples.

4.3.1.4 Module 4: Working with Human-Machine Interfaces

Module 4 will introduce participants to the practical experience of interacting with Human-AI interfaces and system outputs. It will focus on helping end-users understand how to navigate multimodal interface environments, including visual dashboards, alerts, explanation panels, and AR/MR interaction elements.

Participants learn how to input instructions, respond to system requests, read on-screen explanations, and correctly interpret decision prompts within their occupational contexts. The module builds familiarity with the interface: how information will be presented, how feedback mechanisms may appear, and how interface structures are designed to support clarity rather than complexity.

As pilots progress, Module 4 will be enriched by webinar recordings and domain demonstrations, enabling participants to observe real examples of interface use in HumAlne work environments. This

will support confidence and reduce adoption barriers by allowing users to see how others interact with the platform before doing so themselves.

Table 11: Overview of Non-Technical Users' Programme Module 4.

Component	Details
Objective	To familiarise users with the ergonomic, multimodal interfaces developed in HumAlne, including screen-based, visual and AR/MR interfaces.
Topics/Lessons	• HMI navigation and alerts • How to issue and receive commands • Using visual explanations • Understanding feedback responses
Delivery Formats	Primarily delivered through webinars and demonstrations and their recordings (planned for 2026).

4.3.1.5 Module 5: Showcasing Human-AI Collaboration in Diverse Domains

Module 5 connects the organisational relevance of the HumAlne project to participants' own work environments by demonstrating how Human-AI collaboration is applied across various real-world sectors. Through guided case presentations (delivered primarily in interactive webinars and their recordings), users will observe how AI-supported systems operate in different industries such as healthcare, manufacturing, energy, ticketing services, and smart city management.

These domain narratives will illustrate how collaboration methods, trust relationships, system benefits, and ethical considerations differ across sectors. Participants will also see how Human-AI workflows from HumAlne pilots can inspire similar improvements in their own environments, including those external to HumAlne's pilot industries, promoting transferability and wider adoption of collaborative AI practices.

Table 12: Overview of Non-Technical Users' Programme Module 5.

Component	Details
Objective	To showcase transferable Human-AI collaboration concepts demonstrated through HumAlne pilots, inspiring participants to identify opportunities for AI integration in their own work.
Topics/Lessons	• Smart Manufacturing • Smart Cities services (RPA) • Diabetes management • Oncology support • Ticketing • Energy • Beyond HumAlne pilot examples
Delivery Formats	Domain-focused webinars scheduled for 2026; case demonstration recordings; pilot promo material.

4.3.1.6 Module 6: Ethics, AI Regulation and User Rights

Module 6 provides a structured introduction to ethical and legal considerations surrounding the workplace use of AI systems. It explains, in clear language, the principles and objectives of emerging

European AI regulations and the core values underpinning trustworthy AI initiatives—including fairness, safety, transparency, and dignity.

Participants learn what rights they hold as end-users, including their capacity to request explanation, raise concerns, and refuse unsafe instructions. The module also highlights how privacy and data protection obligations apply in the workplace, and why transparent organisational oversight mechanisms protect employees and align AI use with European standards.

Through this training, non-technical users gain assurance that AI deployment is guided by legal and ethical guarantees, increasing confidence and supporting safe long-term uptake.

Table 13: Overview of Non-Technical Users' Programme Module 6.

Component	Details
Objective	To ensure that users understand their digital rights and responsibilities when interacting with AI tools in the workplace.
Topics/Lessons	• Key points of the EU AI Act for end-users • Ethical concerns: fairness, transparency, human dignity • Data protection and privacy basics • How to report ethical or technical concerns
Delivery Formats	Self-paced courses; recorded presentations; slide deck downloads.

4.4 Summary of Training Programme Structure

This section summarized the complete architecture of the HumAlne training offer, detailing two essential learning programmes that cater to both of HumAlne’s macro user-groups: technical and non-technical users. Together, these programmes provide a comprehensive framework for capacity building in Human-AI collaboration, reflecting the project’s technical ambitions, sociotechnical vision and large stakeholder community.

The training programmes combine conceptual, methodological, and practical elements, supporting users who contribute to system development while also accounting for those who engage with AI-driven processes in their daily work. The modular structure ensures flexibility, allowing learners to explore content vertically without their programme or selectively access individual modules according to professional relevance or interest.

The structure presented here has informed the creation of content for the HumAlne Learning Centre and continues to shape an expanded training catalogue via future webinars and the consolidation of future materials (e.g., deliverables, whitepapers, workshop recordings).

5 Expanded Training Catalogue and Content Development

This section documents the current state of the HumAlne Learning Centre training offer at Month 27 and clarifies how content development will continue beyond the end of Task 5.4. As described in Sections 3 and 4, the training offer is now structured into two coherent programmes (**Technical Users** and **Non-Technical Users**), each composed of various modules composed of courses, lessons, and webinars, and other documentation.

Between M18 and M27, the consortium shifted from a fragmented collection of partner resources to a curated catalogue aligned with personas, learning objectives, and platform capabilities. The catalogue now includes full courses, standalone lessons, and recorded or planned webinars, all mapped explicitly to the two training programmes and to the modules defined in Section 4.

The present section is therefore both descriptive and strategic:

Section 5.1 outlines the training catalogue at M27, using structured tables to show how each item is classified and where it sits within the programmes.

Section 5.2 describes ongoing content development, covering lessons and webinars that are under preparation or scheduled and explaining how they will be integrated into the Learning Centre.

5.1 Training Catalogue at M27

At M27, the HumAlne Learning Centre offers a consolidated catalogue of training items grouped under the two training programmes: (1) *“Engineering Human-AI Systems with the HumAlne Platform”* (Technical Users) and (2) *“Empowering Workers in Human-AI Collaboration”* (Non-Technical Users). For each training item, the catalogue specifies:

- **Classification** (course, lesson, webinar)
- **Title / name**
- **Objective**, stated in 1-2 sentences
- **Target audience** (linked to the refined personas in Section 3)
- **Location within the programme** (module and programme)
- **Key components / topics** covered

This classification reflects the expanded scope of T5.4, incorporating and counting the previous courses developed under T5.4 and adding the short, targeted lessons and already-held webinars that have been incorporated. Table 14 summarizes the content.

Table 14: Training Catalogue in HumAlne LC.

Classification	Title	Objective	Location within Training Programmes	Key Components / Topics Covered
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Course	AI Simplified: Everyday Language and Practical Insights	To demystify the main AI mechanisms used in HumAIne, including Active Learning and Explainable AI, so users can interact confidently with the system.	Non-Technical Programme → Module 2	Course overview and study guide outlining Module 2's structure; explains AI in everyday language; clarifies why user feedback matters; introduces how system suggestions are generated; provides study tips, reflection questions, and guidance for self-paced learning.
Lesson	Human-AI Collaboration Essentials – Lecture 1	To build foundational understanding of how humans and AI systems collaborate, and why this approach can enhance work quality and decision-making.	Non-Technical Programme → Module 1	Introductory lesson covering what Human-AI collaboration is, key terms and misconceptions, examples from real work environments (e.g. factories, hospitals, energy control rooms), and an initial view of benefits and risks of AI-assisted work.
Lesson	Human-AI Collaboration Essentials – Lecture 2	To deepen participants' understanding of Human-AI collaboration by reinforcing core concepts with additional examples, benefits/risks, and practical implications for daily work.	Non-Technical Programme → Module 1	Follow-up lesson that revisits core concepts, expands the range of domain examples, addresses typical worries and misconceptions, and highlights how collaboration changes concrete tasks, responsibilities, and communication flows in the workplace.
Course	Introduction to the EU AI Regulation	To ensure that users understand their digital rights and responsibilities when interacting with AI tools in the workplace.	Non-Technical Programme → Module 6	Course overview and study guide introducing key ethics and regulation concepts (fairness, transparency, human dignity, safety), summarising the EU AI Act from an end-user perspective, and explaining how the module is structured (ethics principles, regulatory landscape, accountability and transparency), with prompts for self-paced reflection and learning.

Webinar	Understanding the EU AI Act – Innovation, Governance & Compliance	To explain the EU AI Act in accessible terms, focusing on what it means for innovators, organisations, and users, and how governance, risk, and compliance mechanisms affect the development and use of AI systems.	Non-Technical Programme → Module 6	Overview of the AI Act’s goals and legal nature; definition of an “AI system” and why it matters; risk-based approach (unacceptable, high, limited, minimal risk); timeline and milestones for implementation; governance structure (AI Office, national competent authorities, advisory boards); penalties and enforcement; high-risk systems and conformity assessment; general-purpose AI models; AI Pact and AI Act service desk; AI literacy requirements; regulatory sandboxes and innovation support; discussion of “AI made in Europe” and long-term implications for trust, competitiveness, and responsible AI use.
Lesson	Foundations of Human-AI Collaboration – Lecture 1	Introduce the conceptual foundations of Human-AI collaboration, explaining how human judgement complements technical execution and why human involvement improves AI system performance.	Technical Programme → Module 1	Human vs AI capability asymmetry; Human-in-the-Loop logic; supervision and feedback loops; collaboration patterns; examples of failure prevention.
Lesson	Foundations of Human-AI Collaboration – Lecture 2	Build on the conceptual introduction by exploring deeper interaction mechanisms, transparency pathways, iterative improvement behaviour, and cross-domain benefits of Human-AI collaboration.	Technical Programme → Module 1	Active learning rationale; explainability; incorporating context and domain knowledge; distribution shift; model iteration cycles; pilot-oriented illustrations.

Course	Designing Human-Centric AI Systems with Co-Creative Practises	Provide participants with practical co-creation methods to design, run, analyse, and evaluate Human-AI collaboration workshops and stakeholder sessions aligned with the HumAlne methodology.	Technical Programme → Module 1	Workshop planning and goal setting; facilitation methods; capturing and synthesising outputs; user story creation; risk and KPI mapping; iterative evaluation of sessions.
Course	HumAlne: Hybrid Human-AI Decision Support in Practice	To enable participants to understand the HumAlne concept of hybrid human-AI decision support and apply HumAlne methods, architecture, and tools to design, evaluate, and deploy trustworthy human-centric AI solutions across domains.	Technical Programme → Module 1	Hybrid human-AI decision support; HumAlne learning paradigms (Active, Swarm, Neuro-Symbolic AI); HumAlne architecture and platform; co-creation methodology; trust, ethics, and regulation; pilot use cases; benchmarking human-AI collaboration.
Course	Platform Customisation & Bootstrapping	Train participants in deploying and configuring the HumAlne platform using orchestration pipelines and DevOps approaches within the GFT Factory infrastructure.	Technical Programme → Module 2	GFT Factory environment; Kubernetes orchestration; CI/CD workflow design; RBAC and namespaces; GitLab/Harbor/ArgoCD toolchain; deployment manifests and Helm charts.
Lesson	HumAlne Architecture Lecture	Provide architectural understanding of the HumAlne platform, focusing on its OS-style layered design, workflow orchestration, and integration of human-centric AI paradigms.	Technical Programme → Module 2	HumAlne reference architecture; orchestration and MLOps layers; connectors for Active Learning, Swarm Learning, and NSAI; interoperability; governance and monitoring hooks.
Course	Swarm Learning	Introduce decentralised collaborative learning through Swarm Learning, enabling privacy-preserving, distributed model training without centralised data pooling.	Technical Programme → Module 3	Agent-based local learning; model exchange and merge-and-agree protocols; robustness and privacy advantages; comparison with federated learning; applications in healthcare, IoT, traffic, and fraud detection.

Webinar	Swarm Learning: LLM-Powered Agent-Based Systems in Action	Demonstrate how the Swarm Learning paradigm is combined with LLM-based agents to simplify interaction with the HumAlne platform, enabling users to control MLOps workflows and retrieve experiment insights through a conversational interface.	Technical Programme → Module 3	HumAlne platform context (Kubeflow, Node-RED, MinIO, messaging); swarm-agent architecture; orchestration of specialised LLM agents; chat-based access to pipelines and metrics; use of RAG for project knowledge; self-correcting agent behaviour; demo of usage scenarios.
Webinar	NeuroSymbolic AI Unpacked: What it is and how HumAlne puts it to work	Explain the core ideas of Neuro-Symbolic AI and show how HumAlne's NSAI component combines neural models with symbolic rules, domain expert input, and explainability pipelines to support healthcare decision-making and other high-stakes use cases.	Technical Programme → Module 3	Neuro-Symbolic AI concepts (neural + symbolic reasoning); Logic Tensor Networks and fuzzy rule constraints; domain expert rule definition via UI; LLM-based conversion of natural-language rules; training with domain knowledge embedded in the loss; explanation interface (feature importance + natural-language summaries); diabetes/hypoglycaemia use case; integration with Healthentia and HumAlne architecture; extension with knowledge graphs and graph-RAG for cited, auditable explanations; domain-agnostic applicability in other sectors.
Course	Benchmarking & Performance Evaluation	Present the HumAlne benchmarking suite and train users to evaluate Human-AI collaboration with model-level and human-interaction KPIs.	Technical Programme → Module 5	Multi-layer KPIs (technical, human, system); trust, usability and effort indicators; interaction metrics; diagnostic performance quadrants; dashboard use; fairness and usability assessment.
Course	Core Obligations for AI Developers	Teach participants how to design and govern AI systems in alignment with EU AI Act obligations, applying transparency, documentation, and auditing principles to HumAlne workflows.	Technical Programme → Module 6	Risk-based classification of AI systems; compliance processes; documentation and audit trails; monitoring and oversight; links to fairness and transparency tooling.

Course	Introduction to AI Ethics	Introduce ethical frameworks governing AI development, including fairness, transparency, bias mitigation, and human responsibility requirements.	Technical Programme; Non-Technical Programme → Module 6	Core ethics principles (beneficence, non-maleficence, autonomy, justice); bias and discrimination risks; transparency and explainability; accountability structures; human-in-the-loop responsibility.
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5.2 Ongoing Content Development Beyond M27

Although T5.4 concludes at M27, content creation and consolidation for the HumAIne LC will continue beyond this point. The current catalogue should therefore be understood as a mature but not final configuration. Several additional lessons are currently under development, and thematic webinars planned under T7.4 in 2026 will be progressively integrated. The ongoing content development efforts can be grouped into three main streams, shown in

Table 15: Ongoing content development to be integrated into HumAIne LC.

Additional Lessons for Technical Users	Module 1, Topic 1.4: “The HumAIne vision: positioning the human at the centre” Module 3: Lessons on: • “Neuro-symbolic architectures for reasoning and adaptability.” This will complement the recorded webinar. • “Explainability Techniques (LIME, SHAP, symbolic logic overlays).”
Additional Lessons for Non-Technical Users	Module 3: Lessons on • Reporting problems and unexpected behaviours; • Practical guidelines for safe interaction.
Planned Webinars to be integrated	Pilot-specific webinars, which will be incorporated into Module 7 (technical users) and Module 5 (non-technical users). Benchmarking Suite Standardization in Human-AI Collaboration Explainability Techniques for Human-AI Collaboration

6 Learning Centre: Structure and Navigation

The HumAIne Learning Centre was designed to provide a clear, user-centred entry point into the project’s full training catalogue. Although final course content is still being uploaded and structured, the underlying platform logic, learner pathways, and navigation experience are already established.

When accessing the LC, users will be welcomed with two primary navigation pathways, allowing them to interact with the training material in the format most suitable to their needs and learning objectives.

Training Programme View

This view allows learners to enter the platform following a structured curriculum model. From the main screen, users may select between two options: (1) Technical Users Training Programme and (2) Non-Technical Users Training Programme. Once a programme is selected, the user is taken to a dedicated programme page that displays:

- The modules included in that programme;
- Short programme descriptors;
- The lessons and courses associated with each module within the training programme.

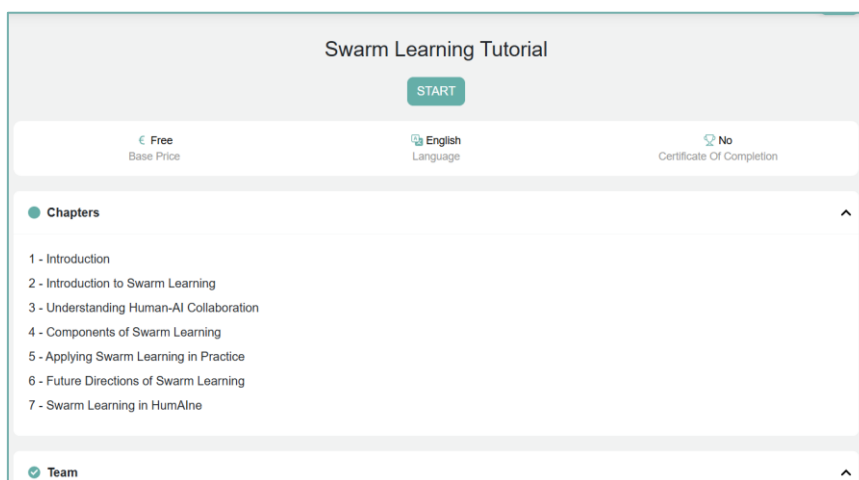


Figure 7: Resource-level information page.

This way, users will be able to see the full offering for their user group and following the programme in the designed order or complete individual lessons according to their needs.

Training Resources View

For users who prefer direct access to individual resources, the Training Resources View provides a complete catalogue of the offered courses and lessons. In this format, each resource appears as a standalone entry and supports topic-based browsing, selective exploration, and easy-to-find lessons.

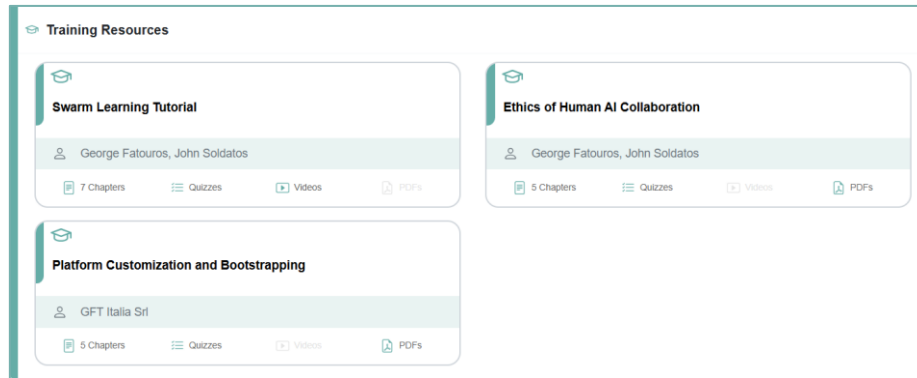


Figure 8: Training resources view of the LC resources.

6.1 Resource Player

Selecting any course or lesson opens a dedicated content player page through which users can:

- Begin exploiting the resource through a “Start” button;
- View content progress through a dynamic completion tracker;
- Read an introductory overview describing the purpose and learning outcomes of the resource;
- Access key conceptual information before starting (definitions, frameworks, terms, etc.);
- Navigate between the different subtopics included in the resource via a navigation sidebar;
- Open linked materials.

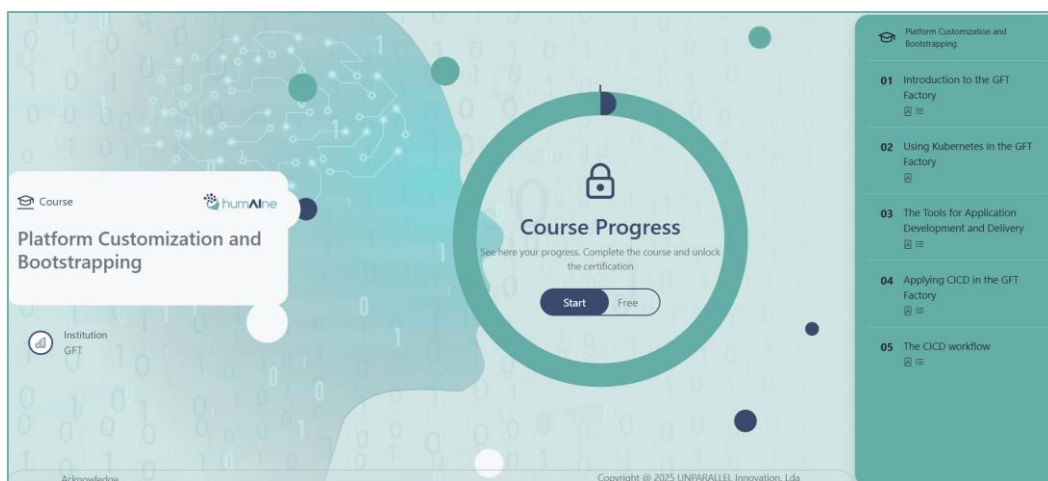


Figure 9: Resource player - main page per training resource.

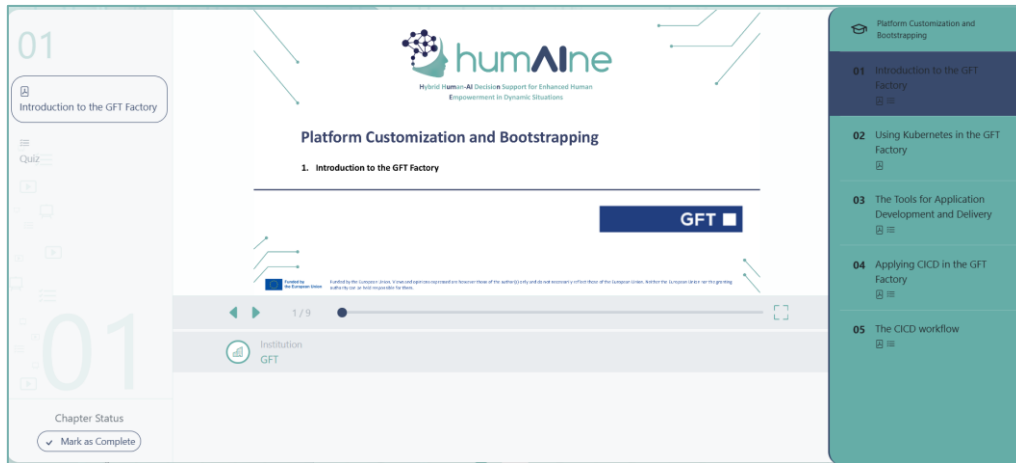


Figure 10: Resource player - subtopic navigation.

This flexible multi-format delivery model supports self-paced learning and accommodates different user preferences.

While the underlying platform is complete and the structure has been established, work continues to upload the developed materials and reformat the LC to follow both view structures.

7 Conclusions

7.1 Summary of Achievements

T5.4 has successfully delivered the core structure, strategy, and training content necessary to support the deployment of the HumAlne Learning Centre. A major outcome is the creation of a substantial portfolio of structured training materials that transform research outputs into reusable learning assets.

To date, the HumAlne training ecosystem comprises:

- **Seven fully defined modules for technical users**, each containing multiple structured learning components.
- **Six fully defined modules for non-technical users**, offering diverse learning tailored to usability, accessibility, and workplace relevance.
- **Nine complete courses** integrated into the two training programmes, combining theoretical material, platform demonstrations, and guided activities.
- **Five standalone lessons incorporated into the training pathways**, providing focused learning resources that address specialised topics.
- **Three webinar recordings**, with additional webinars scheduled for integration in 2026.

Through this scope of work, T5.4 now provides a quantifiable foundation for structured capacity building activities inside and beyond the project. It ensures that HumAlne's scientific, technological, and methodological outputs can be delivered as structured learning experiences rather than isolated documents or reports.

7.2 LC Structure and Internal Validation

The Learning Centre's foundation platform structure has been developed, reviewed internally, and now serves as the central access point for HumAlne training. Internal feedback confirms that the homepage functions effectively as a gateway into all available resources, and that the training content architecture is appropriate for both structured learning programmes and single-resource browsing.

Following the expansion of the training programme from initial module definitions to two complete learning paths (Technical and Non-Technical), the platform structure is now being actively reworked to incorporate these changes. The new structure introduces a **programme-based entry point** where users can select either of the two training pathways, as well as other optimizations for the resource-based view. This redesigned format is being implemented operationally, and once it is complete, users will be able to begin a structured learning programme from the platform home screen, browse all training content through a resource view, navigate lessons and courses through the content players, and progress through materials with completion tracking.

7.3 Post T5.4 Activities

Following T5.4 closure, several key development activities will continue to shape the LC and its training offer. These include:

- Continued content upload and classification, ensuring all existing lessons, courses, videos, and slides are visible and linked to their programme location.

- Integration of platform-based assessment surveys, embedded at the end of each training resource, as a way to measure relevant KPIs and receive continual feedback.
- Integration of pilot-specific and other technical webinars and complementary materials, allowing learners to contextualise HumAlne technology within real operational settings.
- Finalisation of the materials currently under development and integration into the LC.

These activities ensure the LC will remain active and expandable beyond T5.4 without needing structural redevelopment.

7.4 Whitepaper Contribution

A dedicated whitepaper about the HumAlne Learning Centre and its training offer will be produced later in the project to consolidate findings, describe methodology, and position HumAlne's training ecosystem within Europe. The document will aim to summarise the architecture and purpose of the LC, detail how scientific and software outputs were transformed into training assets, and demonstrate how the platform supports capability-building for both user communities. This whitepaper is intended to serve as a transferable reference framework and consolidate all of the final training materials offered in HumAlne.

7.5 Final Remarks

The work completed under T5.4 establishes the technical, pedagogical, and organisational foundations necessary for a sustainable HumAlne Learning Centre. It has transformed scattered project results into an integrated curriculum, formalised two audience-specific training tracks, and implemented a flexible infrastructure allowing training to grow throughout the lifetime of the project and beyond.

The coming phases will focus on full content visibility, usability refinement, and integration of later-stage webinars. Through this layered development approach, HumAlne is positioned to deliver a mature, scalable, and high-impact training environment that will continue strengthening European innovation capacity in human-AI collaboration.